Sub Code:23CS5T02

B. Tech V Semester Regular Examinations **MIC23**

COMPUTER NETWORKS

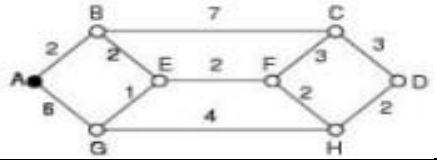
(CSE)

(Model Paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	**What is the primary role of the OSI reference model in networking?**	**2M**	**L1**	**CO1**
	b)	**List the advantages and disadvantages of Mesh Topology?**	**2M**	**L1**	**CO1**
	c)	**Enumerate any four responsibilities of the Data Link Layer in OSI Model?**	**2M**	**L1**	**CO2**
	d)	**What are the different types of Errors that occur in the data in a network ,give Example?**	**2M**	**L1**	**CO2**
	e)	**Briefly Explain the core difference between random access and controlled access protocols?**	**2M**	**L1**	**CO3**
	f)	**Describe the working of pure ALOHA protocol?**	**2M**	**L1**	**CO3**
	g)	**Explain the working of the shortest path routing algorithm?**	**2M**	**L2**	**CO4**
	h)	**What is the count-to-infinity problem in distance vector routing?**	**2M**	**L1**	**CO4**
	i)	**Differentiate between connection-oriented and connectionless transport protocols?**	**2M**	**L1**	**CO5**
	j)	**Explain the structure of a UDP segment?**	**2M**	**L2**	**CO5**
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	**What is a Computer network? Explain the OSI model with diagram?**	**5M**	**L2**	**CO1**
	b)	**Explain Applications of Computer networks?**	**5M**	**L2**	**CO1**
		(OR)			
2.	**a)**	**List the advantages and disadvantages of Bus and Ring Topologies?**	**5M**	**L3**	**CO1**
	b)	**Explain optical fiber cable with the advantages and disadvantages?**	**5M**	**L2**	**CO1**
		UNIT – II			
3.	**a)**	**What is Cyclic Redundancy Check? Calculate CRC code using CRC12 for the following data: 100100,by using the following generator polynomial:$x^{12}+x^{11}+x^3+x^2+x+1$?**	**5M**	**L3**	**CO2**
	b)	**Explain Stop and Wait ARQ protocol with diagram?**	**5M**	**L2**	**CO2**
		(OR)			
4.	**a)**	**Explain how Hamming code can be used for Error detection in computer networks? Calculate 7 bit Hamming code for the following data:1011?**	**5M**	**L3**	**CO2**
	b)	**Explain the services provided by Data link layer?**	**5M**	**L2**	**CO2**
		UNIT – III			
5.	**a)**	**Discuss about CSMA/CA protocol and CSMA/CD protocol?**	**5M**	**L2**	**CO3**

	b)	Explain different types of Controlled Access methods?	5M	L2	CO3
		(OR)			
6.	a)	Explain Standard Ethernet, Fast Ethernet (100 Mbps)?	5M	L3	CO3
	b)	Explain Frequency Division Multiple Access (FDMA)?	5M	L2	CO3
		UNIT – IV			
7.	a)	What is flooding in the context of network routing?What are the advantages and disadvantages of flooding?	5M	L2	CO4
	b)	Find the shortest path from source A to destination D in the given network below using Dijkstra Algorithm? 	5M	L3	CO4
		(OR)			
8.	a)	Explain Leaky bucket algorithm with example?	5M	L3	CO4
	b)	Explain any two Approaches which are followed for Congestion control in a computer network?	5M	L2	CO4
		UNIT – V			
9.	a)	Describe client server communication and HTTP request message format?	5M	L3	CO5
	b)	Explain different services provided by Transport layer?	5M	L3	CO5
		(OR)			
10.	a)	How does TCP ensure reliable data transmission? Explain.	5M	L2	CO5
	b)	Explain Domain Name System(DNS)?	5M	L2	CO5

*** End of the Question Paper ***

Sub Code:23CS5T01 **MIC23**

B. Tech V Semester Regular Examinations

COMPUTER NETWORKS

(CSE)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART – A					**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	**List and briefly explain any three functions of the Network Layer in OSI.**	**2M**	**L1**	**CO1**
	b)	**State two key differences between the OSI Transport Layer and TCP/IP Transport Layer?**	**2M**	**L1**	**CO1**
	c)	**Explain the importance of framing in the Data Link Layer?**	**2M**	**L1**	**CO2**
	d)	**Define flow control and explain why it is essential?**	**2M**	**L1**	**CO2**
	e)	**What is CSMA and how does it reduce collisions?**	**2M**	**L1**	**CO3**
	f)	**Compare FDMA, TDMA, and CDMA in channelization?**	**2M**	**L1**	**CO3**
	g)	**What is hierarchical routing and where is it used?**	**2M**	**L2**	**CO4**
	h)	**What are the general principles of congestion control?**	**2M**	**L1**	**CO4**
	i)	**What are the key features of TCP?**	**2M**	**L1**	**CO5**
	j)	**How does TCP achieve error control?**	**2M**	**L1**	**CO5**
PART – B					**5 X 10 = 50 M**
UNIT – I					
1.	**a)**	**Explain the TCP/IP model with diagram?**	**5M**	**L2**	**CO1**
	b)	**Explain about coaxial cable with the advantages and disadvantages?**	**5M**	**L2**	**CO1**
(OR)					
2.	**a)**	**What is Network topology? Explain different types of network topologies?**	**5M**	**L2**	**CO1**
	b)	**List the differences between LAN,MAN,WAN?**	**5M**	**L3**	**CO1**
UNIT – II					
3.	**a)**	**Explain GO-BACK-N sliding window protocol with diagram?**	**5M**	**L2**	**CO2**
	b)	**Mention different types of Error detection methods ,Explain any two methods with example?**	**5M**	**L3**	**CO2**
(OR)					
4.	**a)**	**What is Burst Error?Find the Checksum for the following data: 10110011101010110101101011010101**	**5M**	**L3**	**CO2**
	b)	**Explain Selective Repeat sliding window protocol with diagram?**	**5M**	**L2**	**CO2**
UNIT – III					
5.	**a)**	**Explain pure ALOHA and Slotted ALOHA with diagrams?**	**5M**	**L2**	**CO3**
	b)	**List the differences between Virtual Circuit and Datagram**	**5M**	**L2**	**CO3**

		Networks?			
(OR)					
6.	**a)**	**Explain Reservation and Polling Controlled Access methods?**	**5M**	**L3**	**CO3**
	b)	**Explain different types of Time Division Multiple Access(TDMA)?**	**5M**	**L2**	**CO3**
UNIT – IV					
7.	**a)**	**Explain Distance Vector Routing in a computer network with Example?**	**5M**	**L3**	**CO4**
	b)	**Explain Token bucket algorithm ?**	**5M**	**L3**	**CO4**
(OR)					
8.	**a)**	**How is tunneling used in connecting incompatible networks?**	**5M**	**L3**	**CO4**
	b)	**Describe the structure of an IPv4 datagram header ?**	**5M**	**L2**	**CO4**
UNIT – V					
9.	**a)**	**Describe the Structure of TCP segment Header ?**	**5M**	**L2**	**CO5**
	b)	**Explain the architectural view of WWW?**	**5M**	**L3**	**CO5**
(OR)					
10.	**a)**	**Describe the architecture of E-Mail and its services?**	**5M**	**L3**	**CO5**
	b)	**Describe HTTP Response message format?**	**5M**	**L3**	**CO5**

Sub Code:23AM5T01 **MIC23**

B. Tech V Semester Regular Examinations

DEEP LEARNING

(AI&ML)

(Model Paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART – A					**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	With a neat sketch of Mc-Culloch Pitts Neuron Model. Explain its salient features.	**2M**	**L2**	**CO1**
	b)	What is Perceptron? Explain the types of Perceptron.	**2M**	**L2**	**CO1**
	c)	What is Feed Forward Neural Network?	**2M**	**L1**	**CO2**
	d)	What are the different Activation Functions used in MLP.	**2M**	**L1**	**CO2**
	e)	What is Drop Connect?	**2M**	**L1**	**CO3**
	f)	What is Batch Normalization?	**2M**	**L1**	**CO3**
	g)	What is the difference between unidirectional and bidirectional RNNs?	**2M**	**L1**	**CO4**
	h)	What are the components of an LSTM Cell?	**2M**	**L1**	**CO4**
	i)	What is meant by generative AI?	**2M**	**L1**	**CO5**
	j)	Name any two applications of Transformers.	**2M**	**L1**	**CO5**
PART – B					**5 X 10 = 50 M**
UNIT – I					
1.	**a)**	Explain about Vector, Scalar, Tensor and Matrices with an example.	**5M**	**L2**	**CO1**
	b)	Explain about Principal Component Analysis.	**5M**	**L2**	**CO1**
(OR)					
2.	**a)**	Explain about Linear Perceptron and its importance in Deep Learning.	**5M**	**L2**	**CO1**
	b)	What is Image and Text Vectorization? Explain briefly.	**5M**	**L2**	**CO1**
UNIT – II					
3.	**a)**	Explain about Multi-Layer Perceptron in Detail.	**5M**	**L2**	**CO2**
	b)	Explain about Back Propagation in Neural Networks.	**5M**	**L2**	**CO2**
(OR)					
4.	**a)**	What is Empirical Risk Minimization?	**5M**	**L1**	**CO2**
	b)	Explain how difficult it is to train Deep Neural Networks.	**5M**	**L2**	**CO2**
UNIT – III					
5.	**a)**	Compare and contrast Adagrad, Adadelta, and RMSprop in terms of learning rate adaptation. In which scenarios would one be preferred over the others?	**5M**	**L4**	**CO3**
	b)	Define the saddle point problem in the context of neural networks. Why is it more problematic than local minima during optimization?	**5M**	**L1**	**CO3**
(OR)					
6.	**a)**	Describe the concept of batch normalization. How does it help in faster and more stable training of deep neural networks?	**5M**	**L2**	**CO3**
	b)	Discuss the need for advanced optimization methods in training deep neural networks.	**5M**	**L6**	**CO3**
UNIT – IV					
7.	**a)**	Describe the architecture of an LSTM unit. Explain the function of each gate.	**5M**	**L2**	**CO4**

	b)	Describe the different types of convolution operations used in CNNs.	**5M**	**L2**	**CO4**
(OR)					
8.	**a)**	Compare and contrast RNNs with Bidirectional RNNs in terms of architecture and applications	**5M**	**L4**	**CO4**
	b)	Explain the need for LSTM over traditional RNNs with respect to long-term dependencies.	**5M**	**L2**	**CO4**
UNIT – V					
9.	**a)**	Describe the role of Transformers in NLP and mention at least two real-world applications.	**5M**	**L2**	**CO5**
	b)	Describe how GPT can be used in speech-related tasks such as transcription or synthesis.	**5M**	**L2**	**CO5**
(OR)					
10.	**a)**	What are the major types of Generative AI models? Briefly describe each.	**5M**	**L2**	**CO5**
	b)	What is Agentic AI? How is it different from traditional AI models?	**5M**	**L1**	**CO5**

***** End of the Question Paper *****

Sub Code:23AM5T01 **MIC23**

B. Tech V Semester Regular Examinations

DEEP LEARNING

(AI&ML)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART – A					**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	What is Deep Learning?	**2M**	**L1**	**CO1**
	b)	What is Vector and Tensor in Deep learning?	**2M**	**L1**	**CO1**
	c)	What is Gradient descent? Explain the variants of Gradient Descent.	**2M**	**L2**	**CO2**
	d)	What is Regularization? Explain their types.	**2M**	**L2**	**CO2**
	e)	What is dropout in Neural network?	**2M**	**L1**	**CO3**
	f)	Explain how to better train the Neural networks.	**2M**	**L2**	**CO3**
	g)	Define convolution operation in CNNs.	**2M**	**L1**	**CO4**
	h)	What is a Restricted Boltzmann Machine (RBM)?	**2M**	**L1**	**CO4**
	i)	What is a variational Auto Encoder?	**2M**	**L1**	**CO5**
	j)	What does GPT stand for in the context of AI Models?	**2M**	**L1**	**CO5**
PART – B					**5 X 10 = 50 M**
UNIT – I					
1.	**a)**	What is Deep Learning? Differentiate between Machine Learning and Deep Learning.	**5M**	**L1**	**CO1**
	b)	Discuss briefly about Matrices, Inverse Matrices and Identity Matrices.	**5M**	**L6**	**CO1**
(OR)					
2.	**a)**	What are Eigen Values and Eigen Vectors? Explain its importance in Deep learning.	**5M**	**L2**	**CO1**
	b)	Explain about Biological Neuron. How Mc-Culloch Pitts neuron model was developed based on Biological Neuron.	**5M**	**L2**	**CO1**
UNIT – II					
3.	**a)**	What is Feed Forward Neural Network? Explain in detail.	**5M**	**L2**	**CO2**
	b)	What is Gradient descent? Explain its importance in neural networks.	**5M**	**L2**	**CO2**
(OR)					
4.	**a)**	With a neat sketch Explain about Deep Neural network and its key aspects.	**5M**	**L2**	**CO2**
	b)	What is Greedy Layer Wise Training? Explain its importance in raining a neural network.	**5M**	**L2**	**CO2**
UNIT – III					
5.	**a)**	What is the role of momentum in optimization, and how does Nesterov Accelerated Gradient (NAG) improve upon classical momentum?	**5M**	**L1**	**CO3**
	b)	Differentiate between dropout and drop connect. How does each method regularize the network and prevent overfitting?	**5M**	**L5**	**CO3**
(OR)					
6.	**a)**	Explain the working principle of the Adam optimizer. What makes it different from other optimization techniques like Adagrad and RMSprop?	**5M**	**L2**	**CO3**
	b)	How do regularization techniques like dropout and batch normalization help in reducing overfitting in neural networks? Support your answer with reasoning.	**5M**	**L1**	**CO3**
UNIT – IV					

7.	**a)**	Describe the architecture of an LSTM unit. Explain the function of each gate.	**5M**	**L2**	**CO4**
	b)	What is a Restricted Boltzmann Machine (RBM)? Explain its architecture and energy function.	**5M**	**L2**	**CO4**
		(OR)			
8.	**a)**	Differentiate between Bidirectional RNNs and Bidirectional LSTMs with diagrams and use-cases.	**5M**	**L5**	**CO4**
	b)	Explain the role of MCMC methods in generative modeling. Why are they necessary?	**5M**	**L2**	**CO4**
		UNIT – V			
9	**a)**	Discuss the architecture and evolution of GPT models (GPT-1 to GPT-4+).	**5M**	**L6**	**CO5**
	b)	Explain how GPT models are applied in: (a) Natural Language Processing (b) Computer Vision *(Give one example for each.)*	**5M**	**L2**	**CO5**
10	**a)**	Explain three real-world applications of Generative AI in creative fields	**5M**	**L2**	**CO5**
	b)	Discuss at least three applications of Agentic AI in real-world systems	**5M**	**L6**	**CO5**

***** End of The Question Paper*****

Sub Code:23CS5T02

B. Tech V Semester Regular Examinations **MIC23**

COMPUTER NETWORKS

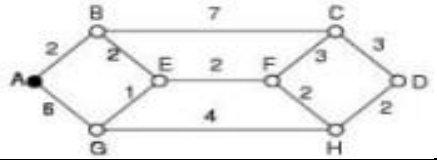
(CSE)

(Model Paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A			**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	**What is the primary role of the OSI reference model in networking?**	**2M**	**L1**	**CO1**
	b)	**List the advantages and disadvantages of Mesh Topology?**	**2M**	**L1**	**CO1**
	c)	**Enumerate any four responsibilities of the Data Link Layer in OSI Model?**	**2M**	**L1**	**CO2**
	d)	**What are the different types of Errors that occur in the data in a network ,give Example?**	**2M**	**L1**	**CO2**
	e)	**Briefly Explain the core difference between random access and controlled access protocols?**	**2M**	**L1**	**CO3**
	f)	**Describe the working of pure ALOHA protocol?**	**2M**	**L1**	**CO3**
	g)	**Explain the working of the shortest path routing algorithm?**	**2M**	**L2**	**CO4**
	h)	**What is the count-to-infinity problem in distance vector routing?**	**2M**	**L1**	**CO4**
	i)	**Differentiate between connection-oriented and connectionless transport protocols?**	**2M**	**L1**	**CO5**
	j)	**Explain the structure of a UDP segment?**	**2M**	**L2**	**CO5**
		PART – B			**5 X 10 = 50 M**
		UNIT – I			
1.	**a)**	**What is a Computer network? Explain the OSI model with diagram?**	**5M**	**L2**	**CO1**
	b)	**Explain Applications of Computer networks?**	**5M**	**L2**	**CO1**
		(OR)			
2.	**a)**	**List the advantages and disadvantages of Bus and Ring Topologies?**	**5M**	**L3**	**CO1**
	b)	**Explain optical fiber cable with the advantages and disadvantages?**	**5M**	**L2**	**CO1**
		UNIT – II			
3.	**a)**	**What is Cyclic Redundancy Check? Calculate CRC code using CRC12 for the following data: 100100,by using the following generator polynomial:$x^{12}+x^{11}+x^3+x^2+x+1$?**	**5M**	**L3**	**CO2**
	b)	**Explain Stop and Wait ARQ protocol with diagram?**	**5M**	**L2**	**CO2**
		(OR)			
4.	**a)**	**Explain how Hamming code can be used for Error detection in computer networks? Calculate 7 bit Hamming code for the following data:1011?**	**5M**	**L3**	**CO2**
	b)	**Explain the services provided by Data link layer?**	**5M**	**L2**	**CO2**
		UNIT – III			
5.	**a)**	**Discuss about CSMA/CA protocol and CSMA/CD protocol?**	**5M**	**L2**	**CO3**

	b)	Explain different types of Controlled Access methods?	5M	L2	CO3
		(OR)			
6.	a)	Explain Standard Ethernet, Fast Ethernet (100 Mbps)?	5M	L3	CO3
	b)	Explain Frequency Division Multiple Access (FDMA)?	5M	L2	CO3
		UNIT – IV			
7.	a)	What is flooding in the context of network routing?What are the advantages and disadvantages of flooding?	5M	L2	CO4
	b)	Find the shortest path from source A to destination D in the given network below using Dijkstra Algorithm? 	5M	L3	CO4
		(OR)			
8.	a)	Explain Leaky bucket algorithm with example?	5M	L3	CO4
	b)	Explain any two Approaches which are followed for Congestion control in a computer network?	5M	L2	CO4
		UNIT – V			
9.	a)	Describe client server communication and HTTP request message format?	5M	L3	CO5
	b)	Explain different services provided by Transport layer?	5M	L3	CO5
		(OR)			
10.	a)	How does TCP ensure reliable data transmission? Explain.	5M	L2	CO5
	b)	Explain Domain Name System(DNS)?	5M	L2	CO5

*** End of the Question Paper ***

Sub Code:23CS5T01 **MIC23**

B. Tech V Semester Regular Examinations

COMPUTER NETWORKS

(CSE)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART – A					**10 X 2 = 20 M**
			Marks	**BTL**	**CO**
I.	**a)**	**List and briefly explain any three functions of the Network Layer in OSI.**	**2M**	**L1**	**CO1**
	b)	**State two key differences between the OSI Transport Layer and TCP/IP Transport Layer?**	**2M**	**L1**	**CO1**
	c)	**Explain the importance of framing in the Data Link Layer?**	**2M**	**L1**	**CO2**
	d)	**Define flow control and explain why it is essential?**	**2M**	**L1**	**CO2**
	e)	**What is CSMA and how does it reduce collisions?**	**2M**	**L1**	**CO3**
	f)	**Compare FDMA, TDMA, and CDMA in channelization?**	**2M**	**L1**	**CO3**
	g)	**What is hierarchical routing and where is it used?**	**2M**	**L2**	**CO4**
	h)	**What are the general principles of congestion control?**	**2M**	**L1**	**CO4**
	i)	**What are the key features of TCP?**	**2M**	**L1**	**CO5**
	j)	**How does TCP achieve error control?**	**2M**	**L1**	**CO5**
PART – B					**5 X 10 = 50 M**
UNIT – I					
1.	**a)**	**Explain the TCP/IP model with diagram?**	**5M**	**L2**	**CO1**
	b)	**Explain about coaxial cable with the advantages and disadvantages?**	**5M**	**L2**	**CO1**
(OR)					
2.	**a)**	**What is Network topology? Explain different types of network topologies?**	**5M**	**L2**	**CO1**
	b)	**List the differences between LAN,MAN,WAN?**	**5M**	**L3**	**CO1**
UNIT – II					
3.	**a)**	**Explain GO-BACK-N sliding window protocol with diagram?**	**5M**	**L2**	**CO2**
	b)	**Mention different types of Error detection methods ,Explain any two methods with example?**	**5M**	**L3**	**CO2**
(OR)					
4.	**a)**	**What is Burst Error?Find the Checksum for the following data: 10110011101010110101101011010101**	**5M**	**L3**	**CO2**
	b)	**Explain Selective Repeat sliding window protocol with diagram?**	**5M**	**L2**	**CO2**
UNIT – III					
5.	**a)**	**Explain pure ALOHA and Slotted ALOHA with diagrams?**	**5M**	**L2**	**CO3**
	b)	**List the differences between Virtual Circuit and Datagram**	**5M**	**L2**	**CO3**

		Networks?			
(OR)					
6.	**a)**	**Explain Reservation and Polling Controlled Access methods?**	**5M**	**L3**	**CO3**
	b)	**Explain different types of Time Division Multiple Access(TDMA)?**	**5M**	**L2**	**CO3**
UNIT – IV					
7.	**a)**	**Explain Distance Vector Routing in a computer network with Example?**	**5M**	**L3**	**CO4**
	b)	**Explain Token bucket algorithm ?**	**5M**	**L3**	**CO4**
(OR)					
8.	**a)**	**How is tunneling used in connecting incompatible networks?**	**5M**	**L3**	**CO4**
	b)	**Describe the structure of an IPv4 datagram header ?**	**5M**	**L2**	**CO4**
UNIT – V					
9.	**a)**	**Describe the Structure of TCP segment Header ?**	**5M**	**L2**	**CO5**
	b)	**Explain the architectural view of WWW?**	**5M**	**L3**	**CO5**
(OR)					
10.	**a)**	**Describe the architecture of E-Mail and its services?**	**5M**	**L3**	**CO5**
	b)	**Describe HTTP Response message format?**	**5M**	**L3**	**CO5**

Sub Code:23CE/EE/ME/EC/CS/IT/AD/AM4T01 **MIC23**

III B. Tech V Semester Regular Examinations

OPERATING SYSTEMS

(AI&ML)

(Model Paper-I)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A	**10 X 2 = 20 M**		
			Marks	**BTL**	**CO**
I.	**a)**	What is an Operating System?	**2M**	**L1**	**CO1**
	b)	What are system calls in operating systems?	**2M**	**L1**	**CO1**
	c)	What is a Process Control Block (PCB)?	**2M**	**L1**	**CO2**
	d)	List any two CPU scheduling algorithms.	**2M**	**L2**	**CO2**
	e)	State Peterson's solution for the critical section problem.	**2M**	**L2**	**CO3**
	f)	Define deadlock in operating systems.	**2M**	**L1**	**CO3**
	g)	What is paging in memory management?	**2M**	**L1**	**CO4**
	h)	Define demand paging.	**2M**	**L1**	**CO4**
	i)	What is a directory structure in file systems?	**2M**	**L1**	**CO5**
	j)	State the goal of protection in an operating system.	**2M**	**L2**	**CO5**
PART – B		**5 X 10 = 50 M**			
		UNIT – I			
1.	**a)**	List and explain any four functions of an operating system.	**5M**	**L2**	**CO1**
	b)	Explain the difference between system programs and application programs	**5M**	**L1**	**CO1**
		(OR)			
2.	**a)**	Write short notes on open-source operating systems.	**5M**	**L2**	**CO1**
	b)	Explain layered and microkernel structures of operating systems with neat sketch.	**5M**	**L2**	**CO1**
		UNIT – II			
3.	**a)**	Describe the role of a dispatcher in CPU scheduling	**5M**	**L2**	**CO2**
	b)	Explain First Come First Serve (FCFS) scheduling with an example.	**5M**	**L3**	**CO2**
		(OR)			
4.	**a)**	Differentiate between preemptive and non-preemptive scheduling.	**5M**	**L4**	**CO2**
	b)	Write short notes on thread libraries.	**5M**	**L**	**CO2**
		UNIT – III			
5.	**a)**	Define semaphores and explain how they are used in process synchronization	**5M**	**L2**	**CO3**
	b)	Write short notes on Peterson's solution.	**5M**	**L2**	**CO3**
		(OR)			
6.	**a)**	Explain the dining philosophers problem	**5M**	**L1**	**CO3**
	b)	What are the necessary conditions for a deadlock to occur?	**5M**	**L2**	**CO3**
		UNIT – IV			
7.	**a)**	Explain the concept of contiguous memory allocation with advantages and disadvantages.	**5M**	**L2**	**CO4**
	b)	What is paging? Draw and explain the structure of a page table	**5M**	**L1**	**CO4**

		(OR)			
8.	**a)**	Differentiate between swapping and paging with suitable examples.	**5M**	**L2**	**CO4**
	b)	Define virtual memory. Explain the concept of demand paging.	**5M**	**L1**	**CO4**
		UNIT – V			
9.	**a)**	What is a file system? Explain different file access methods with examples.	**5M**	**L2**	**CO5**
	b)	Explain the various types of directory structures.	**5M**	**L2**	**CO5**
		(OR)			
10.	**a)**	Describe the file-system operations performed by the operating system.	**5M**	**L2**	**CO5**
	b)	Explain different allocation methods in file systems (contiguous, linked, indexed).	**5M**	**L2**	**CO5**

***** End of the Question Paper *****

Sub Code:23CE/EE/ME/EC/CS/IT/AD/AM4T01 **MIC23**

III B. Tech V Semester Regular Examinations

OPERATING SYSTEMS

(AI&ML)

(Model Paper-II)

Time: 3 hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

		PART – A	**10 X 2 = 20 M**		
			Marks	**BTL**	**CO**
I.	**a)**	List any two functions of an operating system.	**2M**	**L1**	**CO1**
	b)	What is the difference between open-source and proprietary operating systems?	**2M**	**L1**	**CO1**
	c)	Differentiate between process and thread.	**2M**	**L1**	**CO2**
	d)	What is the role of a dispatcher in CPU scheduling?	**2M**	**L2**	**CO2**
	e)	What is a semaphore?	**2M**	**L3**	**CO3**
	f)	Mention two conditions necessary for deadlock to occur.	**2M**	**L2**	**CO3**
	g)	What is thrashing in operating systems?	**2M**	**L2**	**CO4**
	h)	Define swapping in memory management.	**2M**	**L2**	**CO4**
	i)	What is an access matrix in protection?	**2M**	**L2**	**CO5**
	j)	Define file mounting.	**2M**	**L1**	**CO5**
PART – B		**5 X 10 = 50 M**			
		UNIT – I			
1.	**a)**	Define an operating system and explain its main objectives.	**5M**	**L2**	**CO1**
	b)	What are system calls? Explain different types with examples.	**5M**	**L1**	**CO1**
		(OR)			
2.	**a)**	Explain the steps involved in building and booting an operating system.	**5M**	**L3**	**CO1**
	b)	What are the main services provided by an operating system?	**5M**	**L2**	**CO1**
		UNIT – II			
3.	**a)**	Explain the process concept with a neat diagram of process states.	**5M**	**L2**	**CO2**
	b)	What is a Process Control Block (PCB)? List its components	**5M**	**L2**	**CO2**
		(OR)			
4.	**a)**	Write short notes on inter-process communication	**5M**	**L2**	**CO2**
	b)	Explain different multithreading models.	**5M**	**L2**	**CO2**
		UNIT – III			
5.	**a)**	Explain the critical section problem with an example.	**5M**	**L2**	**CO3**
	b)	What are mutex locks? Explain their usage with an example	**5M**	**L3**	**CO3**
		(OR)			
6.	**a)**	Explain the dining philosophers problem.	**5M**	**L2**	**CO3**
	b)	What are the necessary conditions for a deadlock to occur?	**5M**	**L2**	**CO3**
		UNIT – IV			
7.	**a)**	What is copy-on-write? Explain its significance in memory management.	**5M**	**L2**	**CO4**
	b)	Explain the working of page replacement algorithms with an example of FIFO.	**5M**	**L1**	**CO4**
		(OR)			
8.	**a)**	Define thrashing. Why does it occur, and how can it be	**5M**	**L2**	**CO4**

		controlled?			
	b)	Explain different HDD scheduling algorithms with an example (any one algorithm).	**5M**	**L2**	**CO4**
UNIT – V					
9.	**a)**	What is free space management? Explain any two methods with examples.	**5M**	**L1**	**CO5**
	b)	Explain the concepts of file-system mounting and partitions.	**5M**	**L1**	**CO5**
(OR)					
10.	**a)**	Define protection in operating systems. Explain the principles of protection.	**5M**	**L2**	**CO5**
	b)	Explain the access matrix model for implementing protection in file systems.	**5M**	**L2**	**CO5**

***** End of the Question Paper *****

Sub Code: 23MB5T01 **MIC23**

B. Tech V Semester Regular Examinations

ENTREPRENEURSHIP DEVELOPMENT AND VENTURE CREATION

(Common to CSE, AI&DS, AI&ML)

(Model Paper-I)

Time: 3hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART–A					**10X2=20M**
			Marks	**BTL**	**CO**
1.	**a)**	Define entrepreneurship in your own words. What is the core essence of being an entrepreneur?	**2M**	**L1**	**CO1**
	b)	Name two ways in which entrepreneurship contributes to economic development.	**2M**	**L1**	**CO1**
	c)	Define design thinking and name its five stages	**2M**	**L1**	**CO2**
	d)	Explain what a customer persona is	**2M**	**L1**	**CO2**
	e)	What does the "Jobs-to-be-Done" approach emphasize when designing a solution?	**2M**	**L1**	**CO3**
	f)	Define a value proposition.	**2M**	**L1**	**CO3**
	g)	Explain the difference between fixed costs and variable costs	**2M**	**L1**	**CO4**
	h)	What is bootstrapping in the context of startups?	**2M**	**L1**	**CO4**
	i)	What does "scalability" mean in the context of a startup business?	**2M**	**L1**	**CO5**
	j)	Why is storytelling important when pitching a venture to potential investors or partners?	**2M**	**L1**	**CO5**
PART–B					**5X10=50M**
UNIT–I					
1.	**a)**	Illustrate the differences between entrepreneurship and intrapreneurship with examples.	**5M**	**L3**	**CO1**
	b)	Demonstrate with examples how an established company can foster intrapreneurial behavior among employees.	**5M**	**L3**	**CO1**
(OR)					
2.	**a)**	Write about one entrepreneur and explain their journey to success.	**5M**	**L2**	**CO1**
	b)	Identify the main reasons for their success and explain what a young entrepreneur can learn from their story.	**5M**	**L3**	**CO1**
UNIT–II					
3.	**a)**	Identify two macro-level changes happening in India or globally . For each change, explain a startup opportunity that arises.	**5M**	**L2**	**CO2**
	b)	Apply Empathize and Define stages of design thinking to any common problem	**5M**	**L3**	**CO2**
(OR)					
4.		Find two groups of people who struggle to keep a regular fitness routine and explain how the problem is different for each group.	**10M**	**L3**	**CO2**
UNIT–III					
5.	**a)**	Explain what "customer jobs-to-be-done" means and why it is important for designing solutions.	**5M**	**L2**	**CO3**

	b)	Give an example of how understanding customer jobs-to-be-done can create a strong value proposition.	**5M**	**L3**	**CO3**
		(OR)			
6.	**a)**	Explain why knowing the value proposition is important for a business.	**5M**	**L2**	**CO3**
	b)	Define prototyping and MVP. Explain why they are important in product development.	**5M**	**L2**	**CO3**
		UNIT–IV			
7.		Explain the concept of a business model and its types, describe the Lean approach, and outline the nine blocks of the Lean Canvas model with diagram.	**10M**	**L2**	**CO4**
		(OR)			
8.	**a)**	Explain how the lean startup approach to developing a business model differs from a traditional business plan approach	**5M**	**L2**	**CO4**
	b)	Describe how a lean startup handles uncertainties in customer needs or revenue assumptions differently.	**5M**	**L2**	**CO4**
		UNIT–V			
9.	**a)**	Outline a scenario showing how a startup could scale from serving a local market to entering an international market.	**5M**	**L2**	**CO5**
	b)	Analyze the key factors the founders must consider when scaling internationally, such as operations, marketing, and cultural differences.	**5M**	**L4**	**CO5**
		(OR)			
10.	**a)**	Choose a business idea (real or hypothetical) and outline how you would present its scaling potential to an investor.	**5M**	**L6**	**CO5**
	b)	Propose strategies to overcome possible challenges the startup may face during scaling.	**5M**	**L6**	**CO5**

*****End of the Question Paper*****

Sub Code: 23MB5T01 **MIC23**

B. Tech V Semester Regular Examinations

Entrepreneurship Development And Venture Creation

(Common to CSE, AI&DS, AI&ML)

(Model Paper-II)

Time: 3hours **Max Marks: 70M**

Note: 1. Question paper consists of two parts (**Part A and Part B**)
2. Answer all sub questions from **Part A**
3. Answer all questions from **Part B** with **either or choice**

PART–A					**10X2=20M**
			Marks	**BTL**	**CO**
1.	**a)**	How does an intrapreneur differ from an entrepreneur ?	2M	L1	CO1
	b)	Mention one activity you can do on campus to nurture your entrepreneurial skills.	2M	L1	CO1
	c)	Define what is meant by problem validation	2M	L1	CO2
	d)	How does customer segmentation improve the effectiveness of a startup's strategy?	2M	L2	CO2
	e)	Differentiate between a prototype and a minimum viable product (MVP)	2M	L1	CO3
	f)	Why is it beneficial to create low-fidelity prototypes before building a full product?	2M	L1	CO3
	g)	Define Customer Acquisition Cost (CAC)	2M	L1	CO4
	h)	What are components of Business plan?	2M	L1	CO4
	i)	What is Simon Sinek's Golden Circle concept	2M	L1	CO5
	j)	Name three key pieces of information to include in an investor pitch deck.	2M	L1	CO5
PART–B					**5X10=50 M**
UNIT–I					
1.	**a)**	Illustrate the impact of entrepreneurship on society and the economy	5M	L3	CO1
	b)	Demonstrate the potential challenges of high startup activity.	5M	L3	CO1
(OR)					
2.	**a)**	Identify a set of initiatives a college/university could implement to encourage more student entrepreneurship	5M	L3	CO1
	b)	Explain how these initiatives would help students overcome common barriers to starting a venture.	5M	L2	CO1
UNIT–II					
3.	**a)**	Identify the direct competitors and alternatives for the idea of a "smart water bottle that tracks hydration,".	5M	L3	CO2
	b)	Identify and explain the potential challenges your startup might face due to competition.	5M	L2	CO2
(OR)					
4.		Apply the Empathize and Define stages of design thinking to a real-life problem. Describe who you would interview, write a clear problem statement, and explain why it is clear and focused.	10M	L3	CO2
UNIT–III					
5.	**a)**	What are TAM, SAM, and SOM? Give an example for an online learning platform	5M	L2	CO3
	b)	Explain why knowing TAM, SAM, and SOM is important for the startup's strategy and for investors.	5M	L3	CO3
(OR)					

6.	**a)**	Explain how competition analysis and market sizing help in understanding market potential.	5M	L2	CO3
	b)	Apply the iterative design process to show how a product can be improved after poor first user testing results.	5M	L3	CO3
		UNIT–IV			
7.		Choose a startup idea and make a brief go-to-market plan: identify your main customer segment, suggest three ways to reach them, explain why you chose them, and list key metrics to measure success.	10M	L3	CO4
		(OR)			
8.	**a)**	Explain the concept of unit economics and describe how to analyze financial performance using it.	5M	L2	CO4
	b)	Categorize different forms of business organization, identify sources of funds.	5M	L4	CO4
		UNIT–V			
9.	**a)**	Choose a startup idea and write a short pitch including the problem, solution, and what makes it unique.	5M	L3	CO5
	b)	Describe the key components of a compelling entrepreneurial story	5M	L2	CO5
		(OR)			
10.	**a)**	Identify three important things you would look for in a startup pitch before deciding to invest.	5M	L3	CO5
	b)	Explain why each of these things is important for making a good investment decision.	5M	L5	CO5

*****End of the Question Paper*****